

◆ Section 17 — PAC-Driven Control & Execution Physics (Validated)

◆ 17A — PAC-Driven Control (Validated Triadic Control)

17A.1 Introduction

Control in SEKHEM-TSDD is not imposed; it emerges from validated triadic equilibrium.

17A.2 Triadic State

$$X = (R, C, P)$$

17A.3 Triadic Energy

$$E(X) = \|R - C\|^2 + \|C - P\|^2 + \|R - P\|^2$$

E is convex in X and minimized at R=C=P.

17A.4 Validation Gate (from Section 16)

```
\gamma_d =  
\begin{cases}  
1 & \text{if } l_{\mathrm{eff}} \geq \theta_{\mathrm{ind}} \\  
0 & \text{otherwise} \\  
\end{cases}
```

(Use γ_d for the decision gate to avoid confusion with other gains.)

17A.5 Decision Rule

```
D^* =  
\begin{cases}  
\arg\min_X E(X) & \gamma_d = 1 \\  
\text{nothing} & \gamma_d = 0 \\  
\end{cases}
```

17A.6 PAC Correction (Contractive Update)

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\Phi_{\mathrm{PAC}}(X)  
=  
X + \eta(\bar{X} - X), \quad \eta \in (0,1]
```

with $\bar{X} = \frac{1}{3}(R+C+P)$.

Φ_{PAC} is a contraction toward consensus.

17A.7 Control Generation

$$U(t) = \gamma_d \mathcal{H} \big(\Phi_{\mathrm{PAC}}(X(t)) \big)$$

17A.8 Forms of $\mathcal{H}$

- Linear: $\mathcal{H}(Z) = KZ$
 - Nonlinear: $\mathcal{H}(Z) = KZ + \tanh(Z)$
 - Adaptive: $\mathcal{H}(Z, t) = K(t)Z$
-

17A.9 Coupled Dynamics

$$\dot{X} = F(X) + \gamma_d G \big(\Phi_{\mathrm{PAC}}(X) \big)$$

(where G maps PAC state to the state dynamics; keeps notation consistent with $U = \mathcal{H}(\dot{})$).

17A.10 Optimal Control

$$U^*(t) = \arg\min_{\{U\}} \big(E(X) + \lambda J(U) \big)$$
$$J(U) = \|X - Y\|^2 + \beta \|U\|^2$$

17A.11 Differential Control Law

$$\dot{U} = -\nabla_U \big(E + \lambda J(U) \big)$$

17A.12 Multi-Channel Control

$$U = \sum_j v_j U_j, \quad \sum_j v_j = 1, \quad v_j \geq 0$$

17A.13 Theorem — PAC-Induced Stability

Assume:

- $E(X)$ convex and radially unbounded
- $\Phi_{\{\mathrm{PAC}\}}$ is contractive
- $\mathcal{H}$ is bounded/Lipschitz
- $\gamma_d = 1$

Then:

$$\|X(t) - X^*\| \rightarrow 0, \quad \|U(t) - U^*\| \rightarrow 0$$

Sketch: contraction of Φ_{PAC} drives X to consensus; bounded $\mathcal{H}$ preserves dissipativity; combine with Lyapunov $V=E$.

Summary (17A)

Control emerges only under validated triadic equilibrium.

17B — Execution Physics (Validated Physical Layer)

17B.1 Introduction

Execution is a physically realizable process gated by validation.

17B.2 Physical Plant

$M\ddot{q}+D\dot{q}+Kq=U_{\mathrm{real}}(t)$
 with $M>0$, $D\geq 0$, $K\geq 0$.

17B.3 Actuation Mapping

$U_{\mathrm{real}}(t)=\mathcal{A}\big(\gamma_d,U(t)\big)$

17B.4 Constraints

$U_{\mathrm{real}}\leq U_{\max}$

17B.5 Delay

$U_{\mathrm{real}}(t)=U(t-\tau_e),\quad \tau_e\leq \tau_{\max}$

17B.6 Noise

$$U_{\mathrm{real}}(t)=U(t)+\eta(t)$$

17B.7 Multi-Channel Execution

$$U_{\mathrm{real}}=\sum_j v_j U_j$$

17B.8 Execution Dynamics

$$\dot{q}=f(q,U_{\mathrm{real}})$$

17B.9 Execution Condition

$$(I_{\mathrm{eff}}\geq\theta_{\mathrm{ind}})\wedge (E\rightarrow\min)$$

17B.10 Protection Law

$$\gamma_d=0 \rightarrow U_{\mathrm{real}}=0$$

17B.11 Theorem — Physical Stability

Assume:

- (M, D, K) stable (passive/damped)
- $U_{\{\mathrm{real}\}}$ bounded
- τ_e bounded

Then:

$$\|q(t) - q^*\| \rightarrow 0$$

Sketch: standard second-order system with bounded input and damping $\Rightarrow$ asymptotic/ISS stability.

17B.12 Closed Loop

$\text{State} \rightarrow D \rightarrow \gamma_d \rightarrow U \rightarrow U_{\{\mathrm{real}\}} \rightarrow \text{Physics} \rightarrow \text{State}$

◆ 17C — Adversarial & Deception Resistance (Integrated Layer)

17C.1 Principle

Protection is intrinsic to the decision–execution loop.

17C.2 Threat Model

Noise, compromised sensors, hidden correlations, delays, adversarial signals.

17C.3 Core Variable

I_{eff}

17C.4 Rejection Rule

$I_{\mathrm{eff}} < \theta_{\mathrm{ind}}$

$\Rightarrow$

$\gamma_d = 0$

$\Rightarrow$

$U = 0$

17C.5 PAC Robustness

$D = \mathrm{median}(R, C, P)$
Breakdown point $= 33\%$.

17C.6 Hidden Correlation Detection

$\mathrm{Corr}_{\mathrm{hidden}} \uparrow \rightarrow \mathrm{I}_{\mathrm{eff}} \downarrow \rightarrow \gamma_d = 0$

17C.7 Noise Adaptation

$N \uparrow \rightarrow \mathrm{I}_{\mathrm{eff}}^* \downarrow, \theta_{\mathrm{ind}}^* \uparrow$

17C.8 Link to $\mathcal{H}_{\infty}$

Energy attenuation holds (L_2 gain bounded), but

unsafe decisions are pre-emptively rejected.

17C.9 Theorem — Adversarial Rejection

If
 $I_{\mathrm{eff}} < \theta_{\mathrm{ind}}$
or corrupted channel detected
then
 $U(t)=0$

17C.10 Theorem — Safe Execution

If
 $I_{\mathrm{eff}} \geq \theta_{\mathrm{ind}}, \quad \Phi_{\mathrm{PAC}} \text{ bounded}, \quad U \text{ bounded}$
then the system remains stable and safe.

17C.11 Physical Coupling

$\gamma_d=0 \Rightarrow U_{\mathrm{real}}=0$

17C.12 Closed Loop with Protection

$\text{State} \rightarrow D \rightarrow \gamma_d \rightarrow U \rightarrow U_{\{\mathrm{real}\}} \rightarrow \text{Physics}$

17C.13 Final Statement

$\boxed{\text{SEKHEM-TSDD integrates adversarial resistance directly into its decision-execution loop, rejecting any action that fails validated triadic independence.}}$

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
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